

Log In

Input Values

Input Element	Type	Value Specification	Valid	Invalid	Exceptional Cases
Username	String	Must be 5-15 characters long; letters and numbers only	Meets all requirements and has an existing account using it.	The input does not reach the required length, or it is not linked to an existing user account.	Strings with lengths of 0 or 1 as well as extremely long strings and strings that include special characters, control characters, emojis or SQL injection patterns.
Password	String	Must be between 8 and 20 characters and must include at least one letter, one digit, and one special character.	Satisfies the value specification and correctly pairs with the username	Does not meet the length requirement or does not match the username	Passwords of length 0, 1, or extremely long length; passwords with control characters, spaces, or emojis.

Test Case Scenarios

Test Case	Username	Password	Expected Outcome
1	Valid	Valid	The user has successfully logged in.
2	Valid	Invalid	Error: Password is not correct
3	Invalid	Valid	Error: User could not be found
4	Invalid	Invalid	
5	Exceptional	Valid	Error: Incorrect input or an exception was detected.
6	Valid	Exceptional	Error: Incorrect input or an exception was detected.
7	Exceptional	Exceptional	

Concrete Tests

Test Case	Username	Password	Expected Result
1	"rufaelTek"	"abcd1253"	Successful login and redirects to dashboard
2	"rufaelTek"	"xyz"	Error: Password is not correct
3	"rrrrrrr"	"abcd1253"	Error: User could not be found

Log Out

Input Values

Input Element	Type	Value Specification	Valid	Invalid	Exceptional Cases
Current User	String	The provided session ID needs to match an active session ID which the system issued during the login process. The authentication token securely generates token format.	The request contains an active session token that belongs to an authenticated user.	The input fails to match an active session or exists in an incorrect format.	Token is null, empty, expired, or contains invalid symbols such as emojis, control characters, or injection attempts.

Test Case Scenarios

Test Case	Session Token	Expected Outcome
1	Valid	Log out successfully. Redirect to login page
2	Invalid	Error: Session could not be found
3	Exceptional	Error: Invalid or null session

Concrete Tests

Test Case	Session Token	Expected Result
1	"xyz-123-session-token"	Display the successful logout message and reroute to login.
2	"xyz"	Invalid session token error
3	Null	Unauthorized access error

Create Task

Input Values

Note: datetime constructor raises an exception if any parameters are invalid, so datetime data type value specifications are just listed as "must be a valid datetime object."

Input Element	Type	Value Specification	Valid	Invalid	Exceptional Cases
Name	String	Length must be between 1 and 25 characters	Satisfies the value specification and is not used by any tasks belonging to the user.	Does not satisfy the value specification or is already used by a task belonging to the user.	String contains control characters or special characters.
Type	String	Either “To-Do” or “Meeting”	Satisfies the value specification	Does not satisfy the value specification	
Repeat Interval	RepeatInterval	- start_time and end_time (if it exists) are valid datetime objects If end_time exists: - end_time must be a later date than start_time - frequency must be less than or equal to the timedelta between start_time and end_time	Satisfies the value specification	Does not satisfy the value specification	
Duration (Minutes)	Number	- Up to 4 digits - Must be a non-negative number	Satisfies the value specification	Does not satisfy the value specification	
Invitees	String[]	- Must be a list of one or more uid strings - All uids must belong to existing users	Satisfies the value specification	Does not satisfy the value specification	
Priority	Number	Must be within the	Satisfies the	Does not	

		range 1 – 10	value specification	satisfy the value specification	
Due Date	Datetime	- Must be a valid datetime object - Must be a future date	Satisfies the value specification	Does not satisfy the value specification	
Reminders	Boolean	True or False	Satisfies the value specification	Does not satisfy the value specification	

Test Case Scenarios

Test Case	Name	Type	Repeat Interval	Duration (Minutes)	Invitees	Priority	Due Date	Reminders	Expected Outcome
1	Valid	Valid	Valid	Valid	Valid	Valid	Valid	Valid	Show task created confirmation message
2	Valid	Invalid	Valid	Valid	Valid	Valid	Valid	Valid	Show error message
3-38	Valid	Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	
39	Valid	Valid	Invalid	Valid	Valid	Valid	Valid	Valid	Show error message
40-64	Valid	Valid	Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	
65	Valid	Valid	Valid	Invalid	Valid	Valid	Valid	Valid	Show error message
66-81	Valid	Valid	Valid	Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	
82	Valid	Valid	Valid	Valid	Invalid	Valid	Valid	Valid	Show error message
83-91	Valid	Valid	Valid	Valid	Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	
92	Valid	Valid	Valid	Valid	Valid	Invalid	Valid	Valid	Show error message
93-96	Valid	Valid	Valid	Valid	Valid	Invalid	Valid OR Invalid	Valid OR Invalid	
97	Valid	Valid	Valid	Valid	Valid	Valid	Invalid	Valid	Show error message

98-99	Valid	Valid	Valid	Valid	Valid	Valid	Invalid	Valid OR Invalid	
100	Valid	Valid	Valid	Valid	Valid	Valid	Valid	Invalid	Show error message
101	Invalid	Valid	Valid	Valid	Valid	Valid	Valid	Valid	Show error message
102-149	Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	
150	Exception	Valid	Valid	Valid	Valid	Valid	Valid	Valid	Show error message
151-198	Exception	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	Valid OR Invalid	

Concrete Tests

Note: import datetime as dt

Test Case	Name	Type	Repeat Interval	Duration (Minutes)	Invitees	Priority	Due Date	Reminders	Expected Result
1	"Group Meeting"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "Jerry"]	5	dt.datetime(2026, 4, 30, 8)	True	Show task created confirmation message
2	"Group Meeting"	"xyz"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "Jerry"]	5	dt.datetime(2026, 4, 30, 8)	True	Show error message
39	"Group Meeting"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 5, 7, 8), end_time: dt.datetime(1996, 4, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "Jerry"]	5	dt.datetime(2026, 4, 30, 8)	True	Show error message

65	"Group Meeting"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	-1	["Tom", "Jerry"]	5	dt.datetime(2026, 4, 30, 8)	True	Show error message
82	"Group Meeting"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "YZY-YZY-YZXS"]	5	dt.datetime(2026, 4, 30, 8)	True	Show error message
92	"Group Meeting"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "Jerry"]	67	dt.datetime(2026, 4, 30, 8)	True	Show error message
97	"Group Meeting"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "Jerry"]	5	dt.datetime(1996, 4, 30, 8)	True	Show error message
100	"Group Meeting"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "Jerry"]	5	dt.datetime(2026, 4, 30, 8)	NULL	Show error message
101	"	"Meeting"	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency:	60	["Tom", "Jerry"]	5	dt.datetime(2026, 4, 30, 8)	True	Show error message

			dt.timedelta(days: 7)						
150	"\t"	"Meeting "	RepeatInterval(start_time: dt.datetime(2025, 4, 7, 8), end_time: dt.datetime(2025, 5, 7, 8), frequency: dt.timedelta(days: 7)	60	["Tom", "Jerry"]	5	dt.datetime (2026, 4, 30, 8)	True	Show error message

Navigate Dashboard

Input Values

Input Element	Type	Value Specification	Valid	Invalid	Exceptional Cases
Page Request	Page	Must be an existing Page object	Satisfies the value specification	Does not satisfy the value specification	User does not have permission to view requested page (e.g. user timed out)

Test Case Scenarios

Test Case	Page Request	Expected Outcome
1	Valid	Redirect to requested page
2	Invalid	Show error message
3	Exception	Show error message

Concrete Tests

Test Case	Page Request	Expected Result
1	profilePage	Redirect to requested page
2	NULL	Show error message
3	profilePage (user has timed out)	Show error message